X-Ray Diffraction

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1 Abstract

This experiment investigates the fundamental properties of X-ray emission and diffraction using a copper-anode X-ray tube. The lattice constants of Lithium Fluoride (LiF) and Potassium Bromide (KBr) crystals were determined via Bragg diffraction to be 2.019 ± 0.011 Å and 3.277 ± 0.009 Å, respectively. Both values demonstrated strong statistical agreement with accepted literature, deviating by less than 1.5σ. From the minimum wavelength of the Bremsstrahlung spectrum, Planck's constant was subsequently calculated across three varying accelerating voltages. The most precise measurement was obtained at 35 kV, yielding $h = (6.62 \pm 0.21) \times 10^{-34}$ J s, which deviates from the literature value by only 0.03σ. The analysis highlights that higher accelerating voltages significantly improve the signal-to-noise ratio and reduce fitting uncertainty, minimising the ambiguity in selecting the data range.

2 Introduction and Theory

When high-energy electrons strike the copper anode, EM radiation is released through 2 processes, namely bremsstrahlung radiation and characteristic radiation.

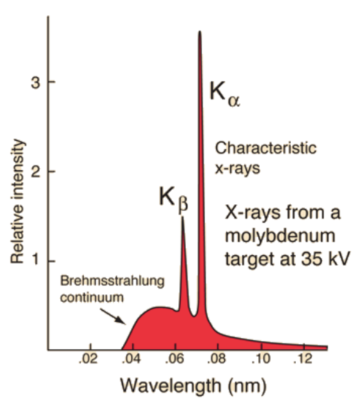


Figure 1: Bremsstrahlung and Characteristic Radiation

2.1 Bremsstrahlung Radiation

Bremsstrahlung, or "braking radiation," constitutes the continuous emission spectrum illustrated in Figure 1. This radiation is produced when a high-energy electron interacts with the electric field of a nucleus rather than colliding with atomic electrons. The attractive Coulomb force between the positively charged nucleus and the negatively charged electron causes the electron to accelerate and deviate from its original trajectory. Because an accelerating charge emits electromagnetic radiation, the electron undergoes a loss in kinetic energy, which is released as an X-ray photon.

The continuous nature of this spectrum arises from the variable acceleration experienced by incident electrons, which is determined by their distance to the nucleus during the interaction. A smaller distance results in higher acceleration and thus the emission of a more energetic photon. In the extreme case where an electron undergoes maximum deceleration (collision with the nucleus), the entire kinetic energy of the electron is converted into a single photon of maximum energy.

2.2 Characteristic Radiation

This type of radiation is responsible for the sharp peaks in Figure 1. When a high-energy electron collides with an electron in the innermost K-shell, that electron becomes ionised and leaves the atom. An energy "hole" is created, which is quickly filled by electrons cascading from higher energy levels.

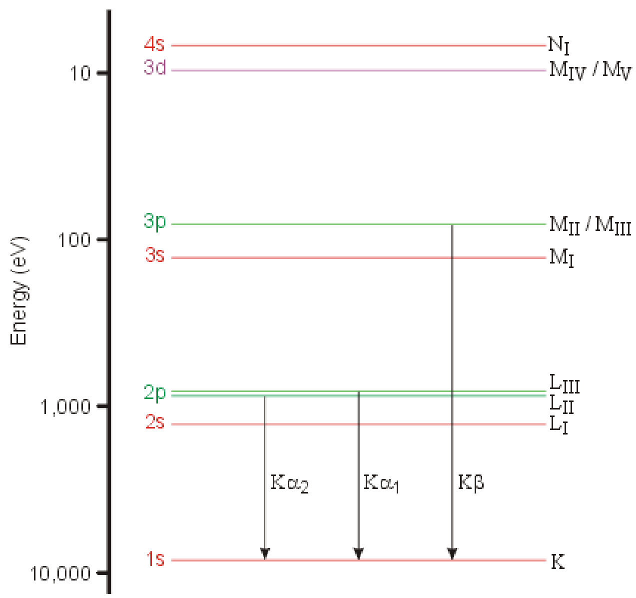


Figure 2: Energy Levels of Copper

These secondary electrons come from the L and M energy levels. The L→K transition and M→K transitions have different energy differences, which leads to the two peaks K_α and K_β. The K_β has a higher energy difference, so it relates to a lower wavelength in Figure 1.

2.3 Bragg Diffraction

X-Ray Diffraction is used to determine the crystal structure of materials by observing how X-rays scatter off atoms to produce interference effects.

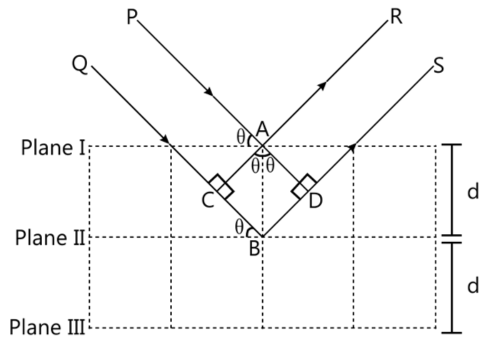


Figure 3: X-Ray Diffraction

The distance $BC = BD = d\sin\theta$, thus the path difference between PAR and QBS is $2d\sin\theta$. When this path difference is equal to integer multiples of the wavelength (λ), constructive inteference occurs which leads to the sharp peaks in the spectrum. This can be represented as:

$$n\lambda = 2d\sin\theta \tag{1}$$

where n is the order of Bragg Diffraction, d is the spacing between the crystal planes, and θ is the angle of incidence.

The purpose of introducing X-Ray Diffraction into this setup is twofold:

1. By varying the angle of incidence, we can allow each corresponding wavelength of radiation to undergo constructive interference and measure the intensity, effectively asking how much of each wavelength is inside the X-rays coming from the anode. This allows us to make calculations using the wavelength, as is required in both bremsstrahlung and characteristic radiation.
2. Verify the literature values of the interplanar spacing for Lithium Fluoride and Potassium Bromide crystals.

2.4 Aims of Experiment

The aims of the experiment are to use X-ray Diffraction to calculate:

1. The lattice constants of LiF and KBr
2. Planck's Constant

3 Experimental Setup

The experiment was conducted using the setup shown below.

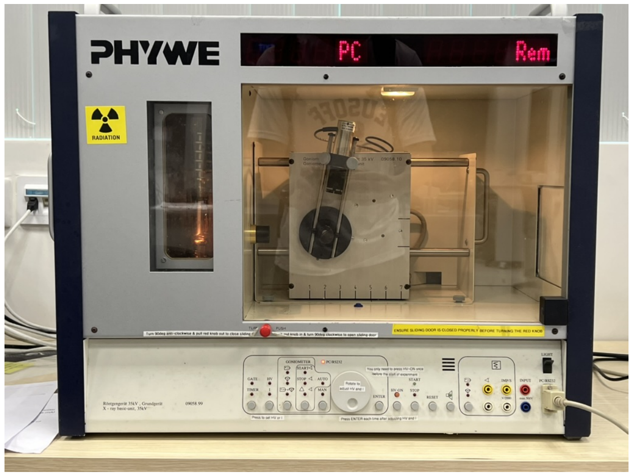


Figure 4: X-ray Diffraction Setup. X-ray tube with copper anode (left) and crystal sample mounted on a goniometer (center).

Experimental parameters, such as the accelerating voltage, and incidence angle, were configured and executed using the PHYWE "Measure" software. This software simultaneously controlled the incidence angle (independent variable) and detected the corresponding count rate (dependent variable), allowing the data to be displayed in real time and exported for analysis.

4 Experiment 1 & 3: Lattice Constant

4.1 Wide Scans

The purpose of the wide scans is to identify the rough location of the peaks, which will then be used to set the ranges of the narrow scans. The angle of incidence was varied from 5° to 55° in increments of 0.5°, with an integral time of 5s. The accelerating voltage was held constant at 35 kV. Current was held constant at 0.05 mA for Experiment 1 (LiF), but this was increased to 0.2mA for Experiment 3 (KBr) as the resolution of peaks was too low at 0.05 mA.

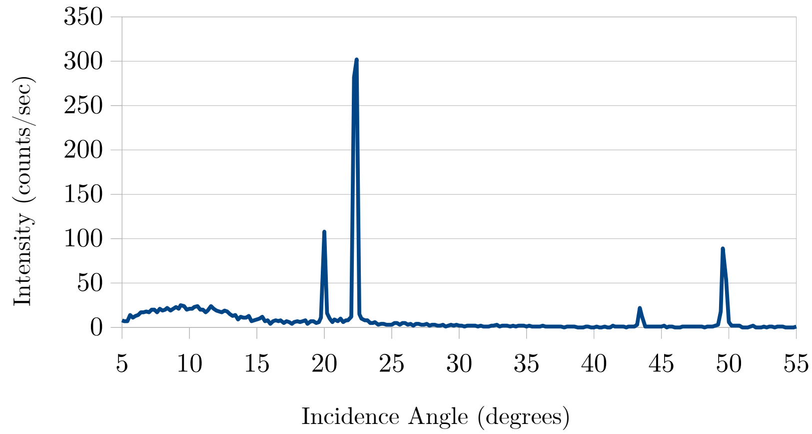


Figure 5: Wide Angular Scan for LiF

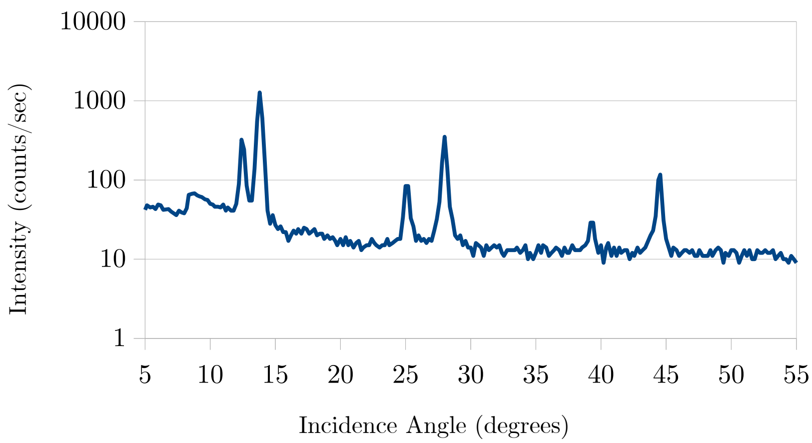


Figure 6: Wide Angular Scan for KBr. Y-axis uses a log scale to compare peaks that differ by 2 orders of magnitude.

The LiF scan shows 4 distinct peaks, while the KBr scan shows 6 peaks. The peaks occur in pairs of K_β followed by K_α. Each subsequent pair arises from higher orders of diffraction, when the path difference of the two reflected rays differ by increasing numbers of integer wavelengths.

4.2 Narrow Scans

Now that the approximate angles of the peaks have been located, a narrow scan is conducted for each peak. The angle of incidence is now varied in increments of 0.1° to improve the resolution of the peak. Integral time has been decreased to 1s, with voltage fixed at 35kV.

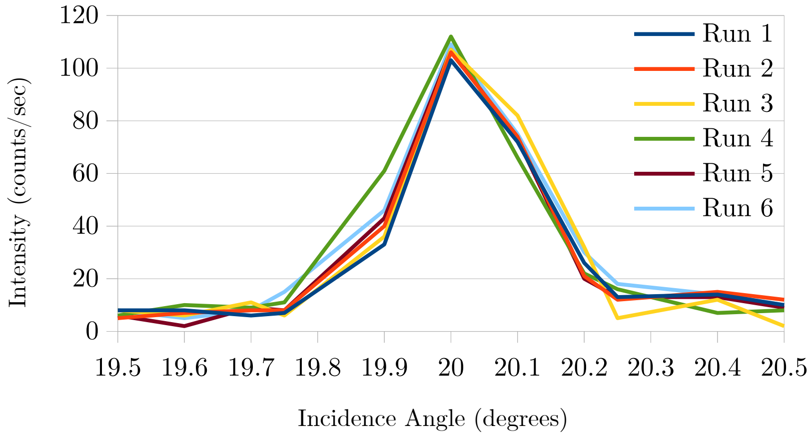


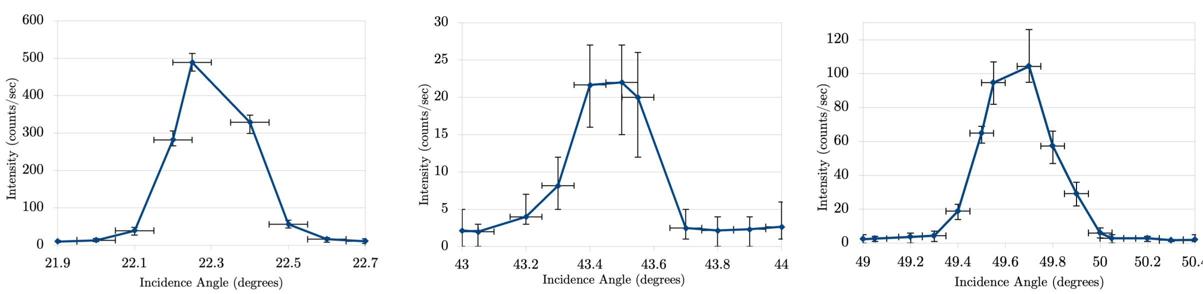
Figure 7: LiF, Peak 1

To compensate for the limited angular resolution of the equipment, the theoretical peak for each experimental run was systematically determined by calculating a weighted average:

$$x_{peak} = \frac{\sum x_i y_i}{\sum y_i} \tag{2}$$

After calculating a distinct x_{peak} for all six runs, the overall mean of these weighted averages was determined, and the corresponding uncertainty ($\Delta\theta$) was evaluated using the sample standard error formula. A full discussion on the combined data analysis will be presented in the subsequent section once all raw data has been introduced.

To thoroughly illustrate the data analysis methodology, all six individual runs were plotted for this initial experiment. However, to maintain visual clarity, the subsequent graphs for Experiments 1 and 3 will plot the average intensity at each incidence angle, utilizing error bars to indicate the maximum and minimum recorded intensities at those respective positions.



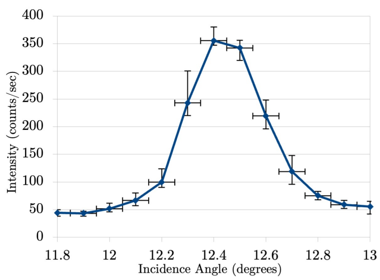
(a) Plot of 2^{nd} Peak

(b) Plot of 3^{rd} Peak

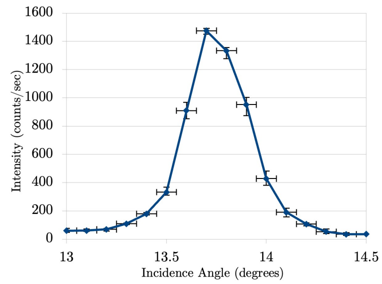
(c) Plot of 4^{th} Peak

Figure 8: Plots of all LiF peaks. Error bars denote minimum and maximum values of intensity obtained for that particular angle, in all 6 runs.

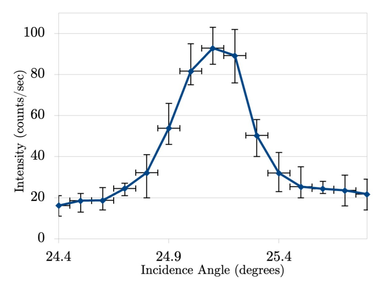
The exact same procedure was repeated for the KBr sample, except for the increased 0.2 mA current to increase the resolution of the peaks.



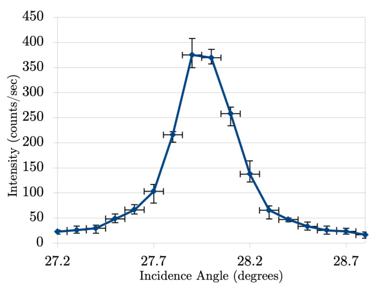
(a) Plot of 1^{st} Peak



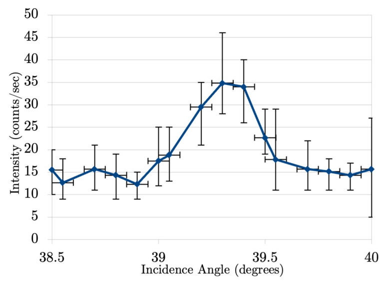
(b) Plot of 2^{nd} Peak



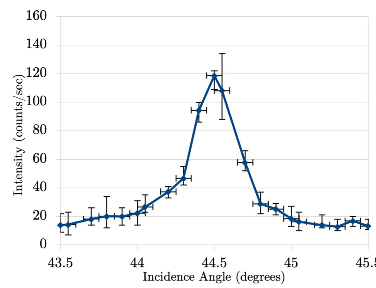
(c) Plot of 3^{rd} Peak



(d) Plot of 4^{th} Peak



(e) Plot of 5^{th} Peak



(f) Plot of 6^{th} Peak

Figure 9: Plots of all KBr peaks. Error bars denote minimum and maximum values of intensity obtained for that particular angle, in all 6 runs.

According to Bragg's Law of diffraction:

$$n\lambda = 2d\sin\theta \tag{3}$$

where n is the order of diffraction, λ is the wavelength of the incident X-rays ($K_\alpha = 1.54$ Å, $K_\beta = 1.38$ Å), d is the crystal lattice constant, and θ is the angle of incidence for each peak ($\theta = x_{peak}$). Plotting $n\lambda$ against $2\sin\theta$ allows the lattice spacing, d, to be determined directly from the gradient of the line of best fit.

To determine this gradient, a least-squares linear regression was applied to the experimental data using the `LINEST` function in Microsoft Excel. The uncertainty of the slope was extracted directly from the regression statistics, representing the standard error of the least-squares fit. The resulting linear fits are plotted in Figure 6 below. While the instrumental uncertainty of 0.05° was formally propagated, it was inconsequential when compared to the magnitude of the statistical uncertainty:

$$\begin{aligned}
\delta d_{total} &= \sqrt{(\delta d_{inst})^2 + (\delta d_{stat})^2} \qquad (4)\\
&= \sqrt{0.0031578^2 + 0.010628^2} \qquad (5)\\
&\approx 0.011\\
&\approx \delta d_{stat}
\end{aligned}$$

and likewise a similar calculation was done for Experiment 3.

The horizontal error bars of magnitude $\delta x = |2\cos\theta| \times \delta\theta$, where $\delta\theta$ is the standard error of the 6 x_{peak} calculations, was omitted from this graph as they were too small to be seen.

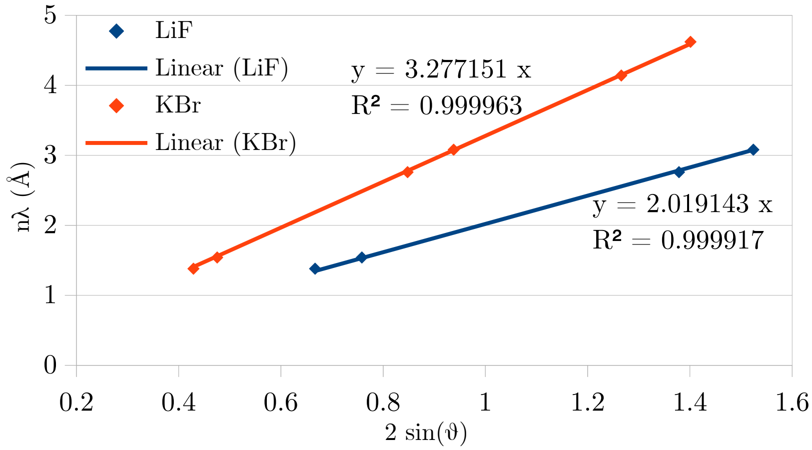


Figure 10: Linear regression of Bragg diffraction data for LiF and KBr crystals. The gradient of each fit corresponds to the lattice spacing d.

From the regression analysis, the experimentally determined lattice spacings are:

Table 1: Experimentally determined lattice spacing (d) for Lithium Fluoride (LiF) and Potassium Bromide (KBr) crystals.

Crystal Type	**Lattice Spacing, $d \pm \delta d$ (Å)**
Lithium Fluoride (LiF)	2.019 ± 0.011
Potassium Bromide (KBr)	3.277 ± 0.009

The high R^2 values validate the Bragg diffraction model for both crystal structures. A comprehensive evaluation of these results, including a comparison to accepted theoretical values and a discussion of potential errors, will be presented in Section 6 (Discussion).

5 Experiment 2: Planck's Constant

5.1 Theory

Through the physical model established in Section 2.1, the governing equations for the radiation can be derived. In this extreme case, the maximum photon energy is equivalent to the total kinetic energy of the electron supplied by the accelerating voltage:

$$E_{max} = hf_{max} = \frac{hc}{\lambda_{min}} = eV \tag{6}$$

This equation serves as the theoretical basis for Experiment 2. By finding λ_{min}, we can substitute in the constants for the charge of a electron (e), the accelerating voltage (V) and the speed of light (c) to calculate experimental values of Planck's Constant (h). The experiment is then repeated for different accelerating voltages of 25 kV, 30 kV and 35 kV.

5.2 Results and Data Analysis

The linear region of the Bremsstrahlung spectrum (for LiF) was visually identified, and a linear least-squares regression was applied exclusively to this data subset. To minimise random error, this regression analysis was performed across six independent runs for each of the three accelerating voltages. The corresponding equations for the lines of best fit are displayed directly adjacent to their respective plots. Only the first experimental run is shown here (for clarity purposes), but analysis was done on all 6 runs.

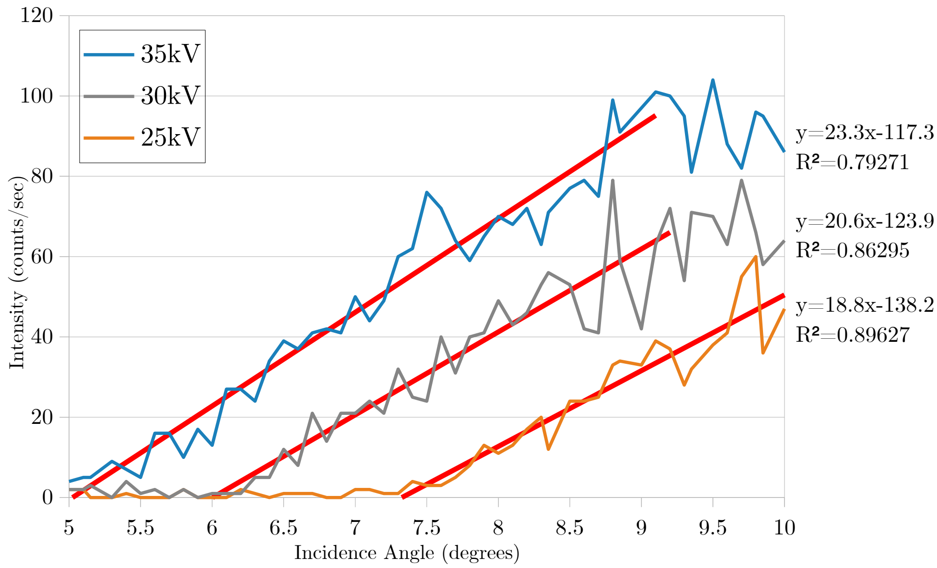


Figure 11: Least squares fit of linear portion of Bremsstrahlung radiation. Endpoints of the red lines denote the boundaries where the spectrum becomes non-linear.

From the linear regression parameters, the minimum diffraction angle is found via the x-intercept:

$$\theta_{min} = -\frac{\text{y-intercept}}{\text{gradient}} \tag{7}$$

Assuming first-order diffraction ($n = 1$), the minimum wavelength is:

$$\lambda_{min} = 2d \sin\theta_{min} \tag{8}$$

The value of Planck's constant for each run (h_i) is calculated using the speed of light (c):

$$h_i = \frac{\lambda_{min} eV}{c} \tag{9}$$

The overall experimental value for a given voltage is the arithmetic mean of the six runs (since all the uncertainties were of roughly the same magnitude).

Table 2: Experimental values of Planck's constant (h) and total uncertainty (δh_{total}) determined at different accelerating voltages.

Accelerating Voltage (kV)	**Planck's Constant, h (10^{-34} J s)**
25	6.77 ± 0.36
30	6.76 ± 0.29
35	6.62 ± 0.21

5.3 Error Analysis

The total uncertainty for each voltage configuration was determined by combining the statistical variance of the repeated runs with the internal fitting error of the linear regression. The lattice constant d was treated as a constant without uncertainty as the fractional uncertainty of d ($\approx$0.0047) was much smaller than the fractional uncertainty of $\sin\theta$ ($\approx$0.05).

Step 1: Uncertainty of the Minimum Angle ($\delta\theta_{min}$)
The minimum diffraction angle, θ_{min}, corresponds to the x-intercept of the linear fit ($\theta_{min} = -c/m$). The uncertainty in this angle was calculated by propagating the standard errors of the gradient (δm) and the y-intercept (δc) obtained from the least-squares regression:

$$\delta\theta_{min} = |\theta_{min}|\sqrt{\left(\frac{\delta c}{c}\right)^2 + \left(\frac{\delta m}{m}\right)^2}$$

Step 2: Fitting Uncertainty per Run (δh_{fit})
Starting with the fundamental equations for the experiment (assuming order of diffraction = 1):

$$\lambda_{min} = 2d\sin\theta_{min} \tag{10}$$

$$h = \frac{\lambda_{min}Ve}{c} \tag{11}$$

Substituting into Equation (10) into (11) gives Planck's constant as a function of θ_{min}:

$$h = \left(\frac{2dVe}{c}\right)\sin\theta_{min} \tag{12}$$

Applying the standard error propagation formula:

$$\delta h = \left|\frac{dh}{d\theta_{min}}\right|\delta\theta_{min} \tag{13}$$

Evaluating the derivative:

$$\frac{dh}{d\theta_{min}} = \left(\frac{2dVe}{c}\right)\cos\theta_{min} \tag{14}$$

Substituting this back into the propagation formula:

$$\delta h = \left|\left(\frac{2dVe}{c}\right)\cos\theta_{min}\right|\delta\theta_{min} \tag{15}$$

Since $\left(\frac{2dVe}{c}\right) = \frac{h}{\sin\theta_{min}}$ from Equation (12), the expression becomes:

$$\delta h = \left|\left(\frac{h}{\sin\theta_{min}}\right)\cos\theta_{min}\right|\delta\theta_{min} \tag{16}$$

$$\delta h_{fit} = |h\cot\theta_{min}|\delta\theta_{min} \tag{17}$$

Step 3: Average Fitting Uncertainty ($\delta h_{fit,ave}$)
Because the reported value of h for a specific accelerating voltage is the mean of six independent runs, the associated fitting uncertainties must be combined.

$$\delta h_{fit,ave} = \frac{1}{6}\sqrt{\sum_{i=1}^{6}(\delta h_{fit,i})^2}$$

Step 4: Total Uncertainty (δh_{total})
Finally, the average fitting uncertainty was combined in quadrature with the statistical standard error of the mean (δh_{stat}) of the six individual h calculations.

$$\delta h_{total} = \sqrt{(\delta h_{stat})^2 + (\delta h_{fit,ave})^2}$$

6 Discussion

6.1 Crystal Lattice Constant

6.1.1 Physical Interpretation

The lattice constant d represents the distance between successive planes in the crystal lattice. Since this value is unique to each crystal, it serves as a structural identifier. This experiment acts as a proof of concept that this structural fingerprint can be determined, proving that an unknown sample could be identified using this method. This constant is influenced by several factors, most notably the atomic radii of the atoms and extrinsic factors like temperature, which results in thermal expansion of the crystal lattice.

6.1.2 Comparison with Literature

Table 3: Summary of experimental and literature lattice constants (d) for LiF and KBr.

Crystal	Experimental d (Å)	Literature d (Å)	% Diff	Statistical Deviation (σ)
LiF	2.019 ± 0.011	2.008	0.55%	1.00
KBr	3.277 ± 0.009	3.290	0.40%	1.44

Both experiments demonstrate high statistical agreement with literature values, as evidenced by the small percentage differences. More importantly, the results deviate from literature by less than 2σ, which places them well within the expected range of experimental error. Therefore, the findings are statistically consistent with theoretical values and should be considered a successful verification of the physical constants.

6.1.3 Sources of Uncertainty

1. **Crystal Imperfections:** Any imperfections in the crystal such as dust, scratches or microscopic cracks would cause the X-rays to scatter unpredictably, leading to additional noise during measurements. This would increase the standard error of the weighted average peak positions.
2. **Low Signal to Noise Ratio:** At certain peaks such as Exp 1 Peak 3 and Exp 3 Peak 5, the intensity becomes comparable in magnitude to the background bremsstrahlung radiation. This creates a low signal-to-noise ratio which increases the uncertainty when calculating the location of peaks. For example, Exp 3 Peak 5 (low intensity) had an x_{peak} uncertainty of 0.016° compared to the 0.003° uncertainty of Exp 3 Peak 2 (high intensity).
3. **Instrumental Uncertainty:** An instrumental uncertainty of half the smallest division was used (0.05°). This contribution was shown to be inconsequential to the final calculated uncertainty.

6.2 Planck's Constant

6.2.1 Physical Interpretation

Planck's constant h acts as the fundamental scaling constant between the energy and frequency of a photon. Thus, this experiment demonstrates the quantised nature of electromagnetic radiation. As for the calculation method, the minimum wavelength (highest frequency) represents the limiting case where the bombarding electron collides with the nucleus and comes to a complete stop, releasing all its kinetic energy as the highest energy photon possible given a fixed accelerating voltage.

6.2.2 Comparison with Literature

Table 4: Summary of experimental results for Planck's constant (h) compared to the literature value of 6.626×10^{-34} J s.

Voltage (kV)	**Exp. $h \pm \delta h$ (10^{-34} J s)**	**% Diff**	**Stat. Dev (σ)**
25	6.77 ± 0.36	2.17%	0.40
30	6.76 ± 0.29	2.02%	0.46
35	6.62 ± 0.21	0.09%	0.03

The experimental results for Planck's constant (h) are all very close to the accepted literature value. All values show a small percentage deviation and fall well within the 2σ threshold for statistical significance. This indicates that the measurements are consistent with the known value of h.

An interesting observation is that the uncertainty of h decreases with increasing accelerating voltages.

- **Improved Signal to Noise Ratio:** Higher accelerating voltages result in a higher intensity of emitted X-rays. This increases the contrast between the zero-intensity region (the pre-cutoff zone) and the rising Bremsstrahlung spectrum. A sharper, more defined "take-off" point reduced the ambiguity when choosing the range of data points.
- **Extensive Linear Region:** As observed in the plotted graphs, higher voltages provide a more prolonged region of linearity before the slope begins to curve. This allows for a larger number of data points to be included in the linear regression analysis. Statistically, increasing the sample size for the fit reduces the standard error of the gradient, and this leads to a more precise final value for Planck's constant.

6.2.3 Sources of Uncertainty, Future Improvements

1. **Subjectivity in Data Ranges:** As the region of linearity was chosen by hand, there is inherent uncertainty in that another person choosing the data range might have chosen slightly differently. Due to the noise of the signal, it is difficult to pinpoint the exact data point where the signal just begins to curve.
   However, this was partially managed by repeating the experiment 6 times and taking the average, which makes it less likely for a single bad choice of data ranges to significantly alter the final result.
   Other methods could also be explored, such as taking the second derivative of the moving average to determine when the graph stops being linear.
2. **Low Signal to Noise Ratio:** Near the cutoff point, low X-ray intensity results in a low signal to noise ratio. This makes the true onset of radiation difficult to distinguish from the background radiation.
   As previously mentioned, as the accelerating voltage increases, the uncertainty decreases. Thus a possible way to improve the experiment would be to conduct it at higher and higher accelerating voltages. Of course, there is also a limit to this, dictated by the voltage at which arcing will start to occur in the tube, as well as temperature constraints. Another alternative is to increase the integral time to reduce statistical noise.

7 Conclusion

The experiment successfully determined the lattice spacings of LiF and KBr crystals using Bragg diffraction. The calculated values were 2.019 ± 0.011 Å for LiF and 3.277 ± 0.009 Å for KBr. Both results are highly accurate and fall well within the experimental error margins of the accepted literature values. This confirms that X-ray diffraction is a reliable method for measuring crystal structures.

The second part of the experiment successfully measured Planck's constant. By analyzing the minimum wavelength of the continuous X-ray spectrum, the most precise value was found at an accelerating voltage of 35 kV, yielding $h = (6.62 \pm 0.21) \times 10^{-34}$ J s. This result is statistically consistent with the accepted theoretical value. The data also confirmed that using higher accelerating voltages reduces total uncertainty because it provides a longer linear region for data fitting and improves the overall signal-to-noise ratio.

The main sources of error in this experiment were the low signal-to-noise ratio at low X-ray intensities and the subjective nature of choosing the linear data range by eye. To improve future experiments, automated curve-fitting software could be used to remove human bias, and longer scanning times could be implemented to reduce random statistical noise.

8 References

1. J.R. Taylor (1997). An Introduction to Error Analysis (2nd ed). University Science Books, Sausalito.